IMPULSE V1: Inferred Microdebris Population from Unmaneuvered LEO State Evolution

Research Supplement

Swayam Sahoo

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Abstract

The sub-10 cm orbital debris environment is poorly constrained. NASA's ORDEM 3.0 and ESA's MASTER 8 diverge by more than two orders of magnitude in key LEO regimes, yet independent observational constraints are scarce. IMPULSE uses decommissioned non-maneuvering LEO satellites as passive dynamical sensors, using historical Two-Line Element (TLE) histories to detect statistically significant orbital perturbations from microdebris impacts. The pipeline uses a 9-state augmented Extended Kalman Filter with Fraser-Potter fixed-interval smoothing, nonsingular equinoctial variables, and per-window quality cuts. Validated through a four-stage synthetic calibration (null test, bias recovery, controlled Δv injection, and clean-versus-debris detection), IMPULSE achieves 50% detection efficiency at $\Delta v_{50} = 0.041\,\mathrm{m/s}$ on synthetic data and $\Delta v_{50} = 0.0385\,\mathrm{m/s}$ on real TLEs. Applied to 3,325 decommissioned spacecraft (15,830 valid analysis windows), the pipeline yields 8,161 candidate events (51.55% trigger rate). After background subtraction via scrambled-residual null tests (19.46% false-positive rate), the net excess is $5{,}080 \pm 103$ events, or 32.09% per window. This excess dwarfs the debris predictions: ORDEM predicts 847.9 detectable events (5.36%), while MASTER predicts only 6.1 (0.04%). The implied non-debris systematic floor is 26–32%, preventing ORDEM/MASTER discrimination under the baseline operating regime. This result establishes the *information limit* of TLE-based passive sensing under the baseline operating regime and charts a V2 path toward breaking the degeneracy.

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1 Motivation: The Microdebris Observational Gap

Conventional space surveillance cannot resolve individual debris objects below $\sim$10 cm. Because this population must be characterized statistically, engineering models like NASA's ORDEM 3.0 and ESA's MASTER 8 are essential. But they are built from different observational inputs and fragmentation physics, and they disagree substantially. In critical regions, they can disagree by factors of over $\sim$100$\times$.

Independent validation is a difficult task, and therefore scarce. Returned surfaces give small data points, and dedicated impact detectors like DEBIE and MEDI are trajectory-specific. Ground based surveys are geographically limited.

IMPULSE proposes an alternative observational approach. Decommissioned LEO spacecraft are essentially naturally occurring test particles. They have no reason to maneuver, their orbits are continuously tracked via TLEs, and they sample the debris belt for years at a time. The key question is if TLE precision, along with accurate dynamical modeling, is sufficient to detect subtle orbital perturbations

caused by centimeter-millimeter scale debris impacts, and whether a fleet of them can discriminate between ORDEM and MASTER.

2 The IMPULSE Measurement System

2.1 Core Idea

IMPULSE analyzes each spacecraft's TLE history in 90 day windows. For each window, a forward Extended Kalman Filter (EKF) is run to estimate orbital state and unmodeled acceleration. Then, from the terminal state, a backward EKF runs in reverse. Finally, a Fraser-Potter fixed-interval smoother fuses the two into a retrospective estimate. The statistic used for detection is the Mahalanobis distance between the forward and backward state estimates, normalized by the smoother covariance. A sharp peak here indicates a localized inconsistency in the dynamical mode, a candidate for a debris strike.

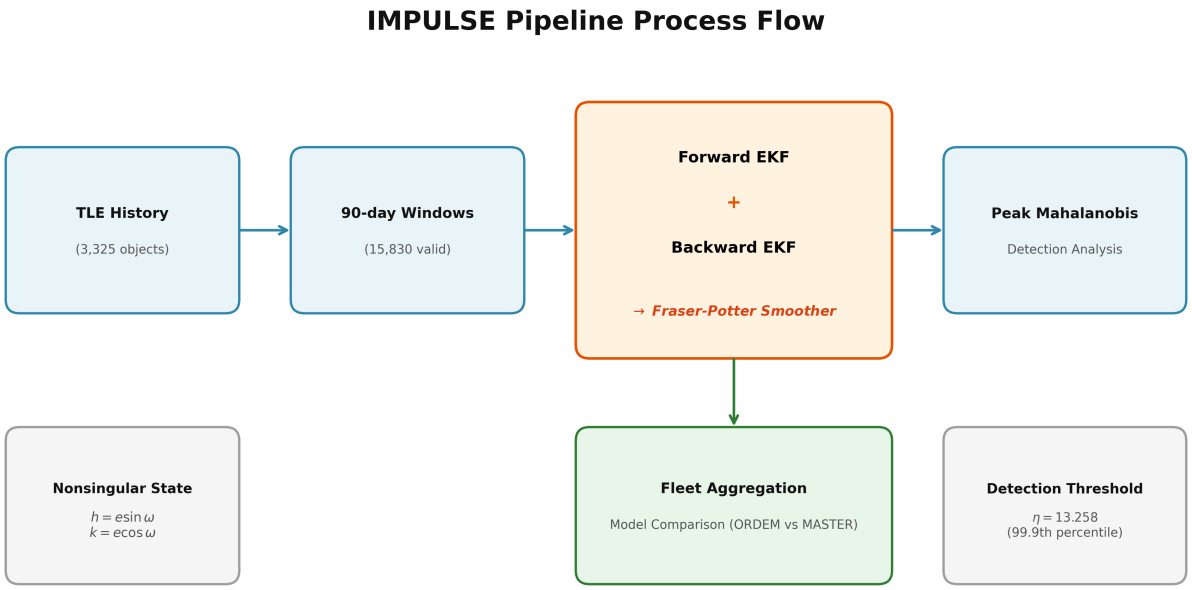


Figure 1: The IMPULSE pipeline. TLE histories for 3,325 objects are segmented into 90-day windows. Each window is processed by a forward EKF, backward EKF, and Fraser–Potter smoother. The peak Mahalanobis distance is tested against a null-calibrated threshold ($\eta = 13.258$, 99.97th percentile). Detections are aggregated across the fleet for model comparison.

2.2 Methodological Design

Nonsingular state representation. In the Gauss planetary equations for $d\omega/dt$, there is a $1/e$ singularity. As eccentricity approaches zero due to drag-driven circularization, the $1/e$ term diverges, causing filter instability. IMPULSE eliminates the singularity by using equinoctial style substitution. $h = e\sin\omega$ and $j = e\cos\omega$. The resulting measurement model is a direct readout of the first six states, while the Jacobian singularity is avoided.

Force model. The null model includes J_2 to J_4 harmonics, atmosphere drag with stochastic C_d drift following an Ornstein-Uhlenbeck Process, solar radiation pressure modeling, and third body perturbations (lunisolar).

Quality cuts. Before filtering, each 90 day window goes through quality cuts. ≥ 15 TLEs, no gaps > 14 days, altitude drop < 10 km, and exclusion of the first window for objects with over 90 days of data. After the cuts, 69.2% of the windows (15,830 out of 22,888 total) remain valid.

3 Validation Ladder

IMPULSE goes through a validation ladder with each stage testing a more demanding property of the system.

Gate	Question	Result
1. Null Test	Does the filter trigger under ideal conditions?	99.97th pct threshold $\eta = 13.258$
2. Bias Recovery	Can it track slowly varying forces?	Recovers 10^{-6} m/s^2 bias
3. Strike Injection	Can it recover known impulses?	$\Delta v_{50} = 0.041$ m/s (synthetic)
3b. Real-TLE Injection	Does sensitivity persist on real TLEs?	$\Delta v_{50} = 0.0385$ m/s
4. Clean vs. Debris	Does it distinguish a realistic debris population?	Mean peak Maha 7.39 vs. 6.51

Table 1: Validation ladder. Each gate tests a specific property of the measurement system under progressively more realistic conditions.

Gate 3 (Synthetic Strike Injection). Synthetic Δv strikes are injected along-track across a logarithmic grid (10^{-4} to 0.5 m/s). The resulting efficiency curve rises sharply between 10^{-2} and 10^{-1} m/s, which is the pipeline's sensitivity transition.

Gate 3b (Real-TLE Strike Injection). Same impulses were injected into real TLE windows to test if this sensitivity persists under observational noise and real histories. It revealed a $\sim 15\%$ trigger rate at negligible impacts (10^{-4} m/s), which represents the probability that a real TLE window contains an artificial peak within the ± 1-day recovery interval.

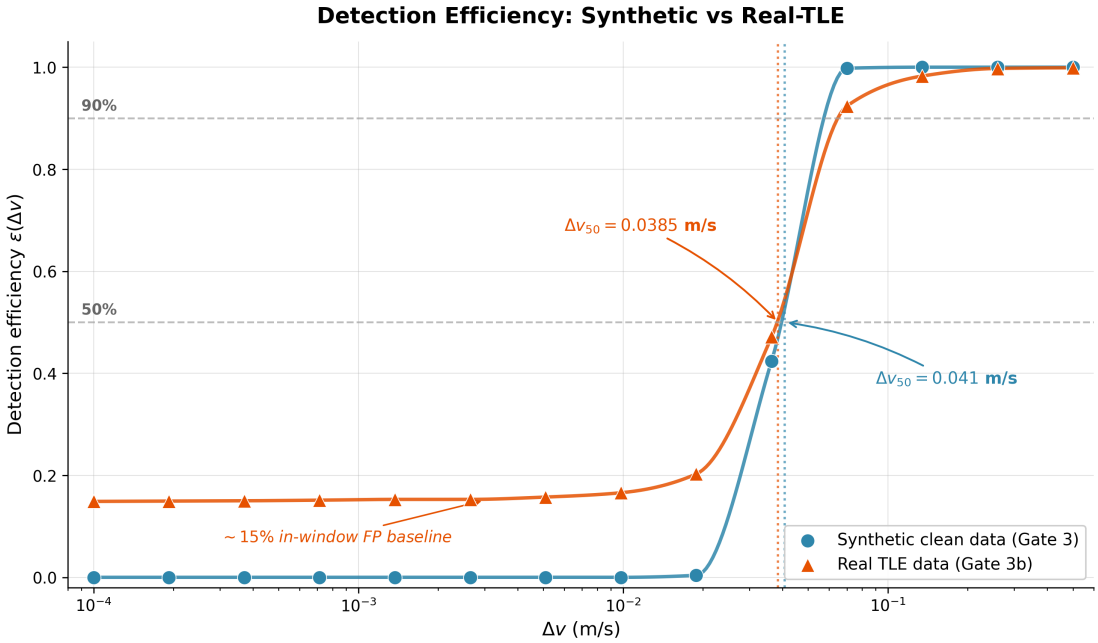


Figure 2: Detection efficiency $\epsilon(\Delta v)$ for synthetic clean data (blue circles, Gate 3) and real-TLE injection (orange triangles, Gate 3b). The real-TLE $\Delta v_{50} = 0.0385$ m/s is slightly better than the synthetic $\Delta v_{50} = 0.041$ m/s consistent with quality-cut pre-filtering. The $\sim$15% baseline at low Δv on real TLEs is the in-window false-positive rate from noise.

4 Fleet-Scale Application

4.1 The Fleet

The real fleet comprises 3,325 decommissioned LEO spacecraft with continuous TLE coverage from 2009-2026. The altitude distribution peaks in the sun-synchronous region (700–1000 km), with inclinations spanning the full LEO band

4.2 Results

The fleet was processed via memory-safe multiprocessing. Summary results:

Metric	**Value**
Total objects processed	3,325
Valid analysis windows	15,830
Peaks computed	15,786
Mean peak Mahalanobis	19.44
99.97th percentile threshold	13.258
Raw detections ($>\eta$)	**8,161**
Raw trigger rate	**51.55%**

Table 2: Fleet-scale detection summary. The raw trigger rate of 51.55% far exceeds the ∼0.06% expected under the ideal synthetic null.

The raw trigger rate of 51.55% is important. Over half of the windows exceeded the 99.97th percentile threshold. This suggests that real TLEs contain temporally coherent structures not accounted for in synthetic null tests.

4.3 Background Subtraction

The synthetic null rate (0.06%) does not apply to real TLEs. It is a reflection of the model's intrinsic false-positive rate under ideal conditions. The empirical false-positive rate was measured via scrambled-residual null tests. A smooth spline was fit to each element's time series, residuals then randomly permuted, and the same pipeline was run on the scrambled data. This preserves the noise distribution and destroys temporal coherence.

Metric	**Value**
Valid windows	15,835
Triggers ($>\eta$)	3,081
Scrambled FP rate	**19.46%**

Table 3: Scrambled-residual null test results. The 19.46% rate is the empirical false-positive background against which real detections must be compared.

The background-subtracted net excess is:

$$N_{\text{net}} = N_{\text{obs}} - N_{\text{FP}} = 8{,}161 - 3{,}081 = 5{,}080 \pm 103 \quad (32.09\% \text{ per window}). \tag{1}$$

The net excess is a primary result. It represents all temporally coherent statistically significant orbital perturbations. This includes debris impacts, but also confounders like unmodeled maneuvers, catalog artifacts and geomagnetic storm residuals.

5 Model Comparison

ORDEM and MASTER predicted detectable impacts were computed with Monte Carlo sampling over the fragment population, convolved with the efficiency curve $\epsilon(\Delta v)$, and weighted by per-object observation duration and cross sectional area ($A = 1.0\,\mathrm{m}^2$, $m = 200\,\mathrm{kg}$).

Model	Expected Impacts	Predicted Detected	Effective ϵ
ORDEM	131,997	847.9	0.64%
MASTER	968.9	6.1	0.63%

Table 4: Model predictions propagated through the IMPULSE selection function. The ORDEM/MASTER ratio remains 138× even after detection efficiency degradation.

ORDEM/MASTER ratio: 138×. Even after degrading through the detection efficiency filter, the models predict vastly different numbers of detectable events.

5.1 The Discrimination Test

The pre-specified decision criterion requires $|\log_{10} \mathcal{B}| > 1$ for decisive model discrimination. Comparing the observed net excess against the predictions:

Scenario	Predicted Total	Observed	Residual
ORDEM + scrambled FP	$3{,}081 + 848 = 3{,}929$	8,161	+4,232
MASTER + scrambled FP	$3{,}081 + 6 = 3{,}087$	8,161	+5,074

Table 5: Model comparison against observations. Both scenarios underpredict by a factor of ∼2. The residual is the non-debris systematic floor.

Both scenarios under-predict the observation by a factor of 2. The residual excess (ORDEM: 4,232, MASTER: 5,074) represents the non-debris temporally coherent systematic TLE noise.

Model	Debris Rate	Implied Systematic	Net Excess
ORDEM	5.36%	26.73%	32.09%
MASTER	0.04%	32.05%	32.09%

Table 6: Decomposition of the 32.09% net excess. The systematic floor is 5–6× larger than the ORDEM debris signal and 800× larger than the MASTER signal.

The systematic floor is 5-6 times larger than the ORDEM debris signal and 800 times larger than the MASTER signal. The debris difference of 5.3 percentage points is smaller than the systematic uncertainty (26-32 percentage points). The Bayes factor falls squarely in the indifference zone $[-1, +1]$. The IMPULSE pipeline as V1 cannot discriminate ORDEM from MASTER.

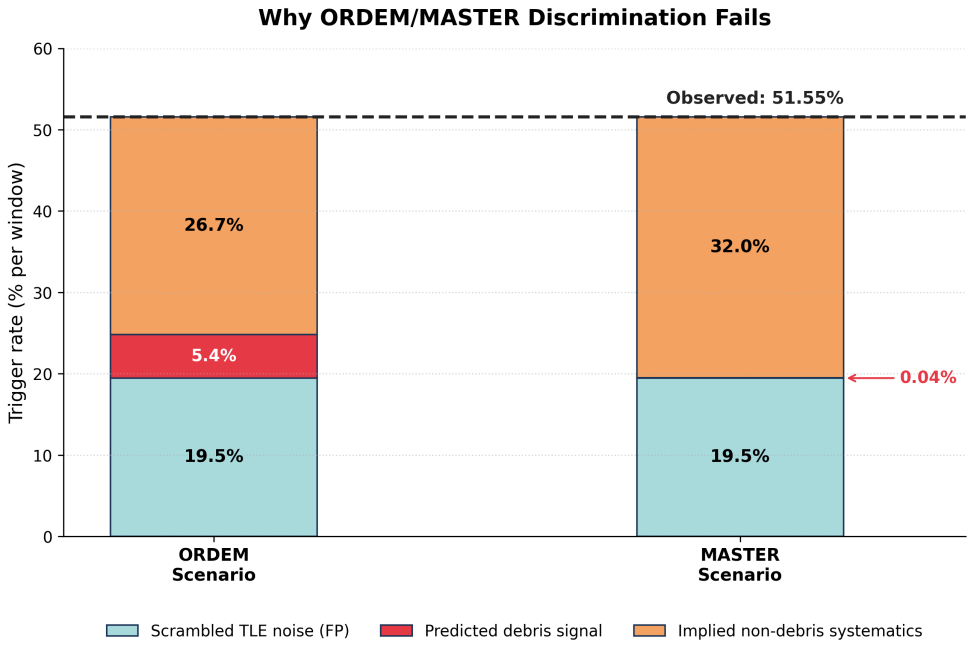


Figure 3: Decomposition of the observed 51.55% trigger rate. The scrambled TLE noise contributes 19.5% (measured). Under the ORDEM scenario, debris contributes 5.4% and the implied systematic floor is 26.7%. Under MASTER, debris is 0.04% and the implied systematic is 32.0%. In both cases the systematic floor dwarfs the debris signal, blocking model discrimination.

This is not a failure of the approach or experiment. It is a measurement of the **information limit** of TLE-based passive sensing under the **baseline** operating regime. We have established that:

1. The detection pipeline is validated, calibrated, and performs as designed.
2. The real-TLE temporally coherent non-debris systematic floor is 26–32% per window.
3. The ORDEM debris signal is ∼5.4%, the MASTER signal ∼0.04%.
4. The signal-to-systematic ratio is < 0.2 for ORDEM and < 0.002 for MASTER.

6 The V2 Path: Decomposing the Non-Debris Systematic Floor

V1 establishes a population-level excess of triggers, but these cannot yet be uniquely attributed to debris. The observed rate can be decomposed as

$$N_{\text{obs}} = N_{\text{TLE,FP}} + N_{\text{nondebris,coherent}} + N_{\text{debris}}, \tag{2}$$

where $N_{\text{TLE,FP}}$ represents false positives from TLE noise and sampling, and $N_{\text{nondebris,coherent}}$ represents temporally coherent systematic signals unrelated to debris. After accounting for the measured TLE false-positive floor, the remaining excess is approximately 26–32%. Discriminating ORDEM from MASTER therefore requires decomposing this excess and constraining the non-debris contribution.

The most promising avenue to explore first is **object-type stratification**. Rocket bodies are inert and rarely maneuver compared to payloads which may have additional post-decommision activity. Differences between these can therefore constrain the contribution of object-specific systematics to the excess.

Additionally, correlations with geomagnetic activity (K_p), cross-referencing public maneuver records and temporally monitoring excess rates over the solar cycle are more routes that V2 will explore. The

objective is to reduce the systematic floor enough that the remaining excess can be compared against the substantially different debris populations predicted by ORDEM and MASTER.

7 What V1 Establishes

IMPULSE V1 demonstrates that TLE histories contain a substantial population of temporally coherent orbital anomalies, but the baseline measurement system cannot uniquely attribute them to debris. The measured non-debris systematic floor of 26–32% per window overwhelms the predicted debris signals from ORDEM (5.36%) and MASTER (0.04%), preventing model discrimination.

This establishes the **information limit** of the baseline TLE-based approach: the measurement system is sensitive to orbital perturbations, but the systematic floor must be reduced before the debris signal becomes observable. V2 will address this through systematic decomposition and object-type stratification using the existing V1 dataset.

This supplement reflects the completed IMPULSE V1 analysis as of September 4, 2026.